



What does a medicine's price **pay for?**

Part 1 showed the active ingredient became cheaper over four decades. This part reviews the rest of the cost structure — **eight companies' SEC filings** (SEC: the US Securities and Exchange Commission, the regulator every listed company reports to), human pharma and animal health side by side. The difference explains a twenty-year price divergence.

One industry code, **two cost structures.**

R&D • SHARE OF REVENUE

19% vs 7%

Human pharma median **19.4%**; animal health **7.3%** — a **2.7x** difference in the share of revenue reinvested in R&D.

COST OF GOODS • SHARE OF REVENUE

30% vs 45%

The ranking inverts: human **29.6%**, animal **44.6%**. Animal health spends the larger share on manufacturing, where the ingredient sits.

WHAT REMAINS AFTER COGS+R&D+SG&A

~30 vs ~23

Per \$100 of revenue: human ~30, animal ~23. **Similar remainders, reached through different structures.**

This is consistent with the price record: an R&D-weighted cost structure finances future products through price — human prices rose +75% real over twenty years. A manufacturing-weighted structure follows its input costs — veterinary prices fell 4% real over the same window. **The cost structure and the price behaviour agree.**

Straight from the filings. Nothing modelled.



Source

Annual 10-K figures via **SEC EDGAR's XBRL API** (SEC: US Securities and Exchange Commission; the 10-K is the audited annual report every US-listed company must file) — machine-readable, public, keyless.



Measure

COGS, R&D and SG&A as % of revenue — **mean of FY2023-25** per company, **median** across each group so no single filer drags the figure.



Two clean groups

Human: Pfizer, Lilly, AbbVie, Bristol Myers Squibb. **Animal:** Zoetis, Elanco, Phibro — pure-play filers only. Merck (~9% animal health) is shown but **pooled with neither group**. Animal-health bars are shown **unattributed**: the finding is the group contrast, not any one filer's margin. All figures re-derivable from the filings above.

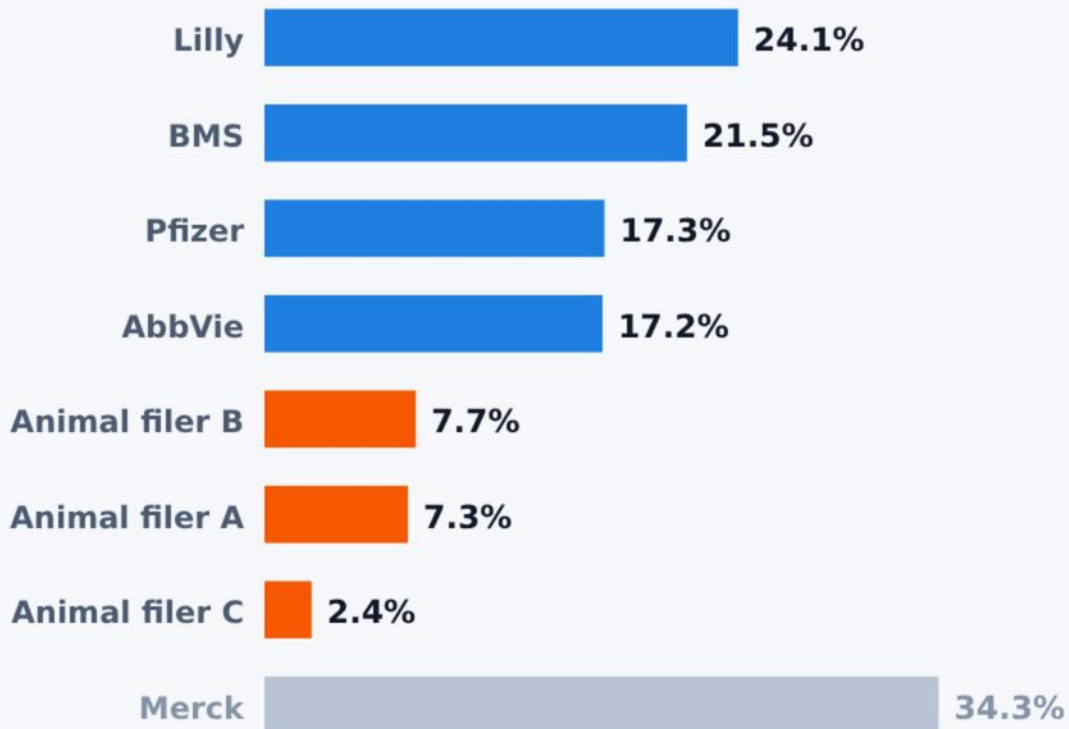


What COGS is not

COGS is not the ingredient. No filer splits API out of its cost of goods — the ingredient from Part 1 sits *inside* these bars, unquantified.

R&D share: the main difference.

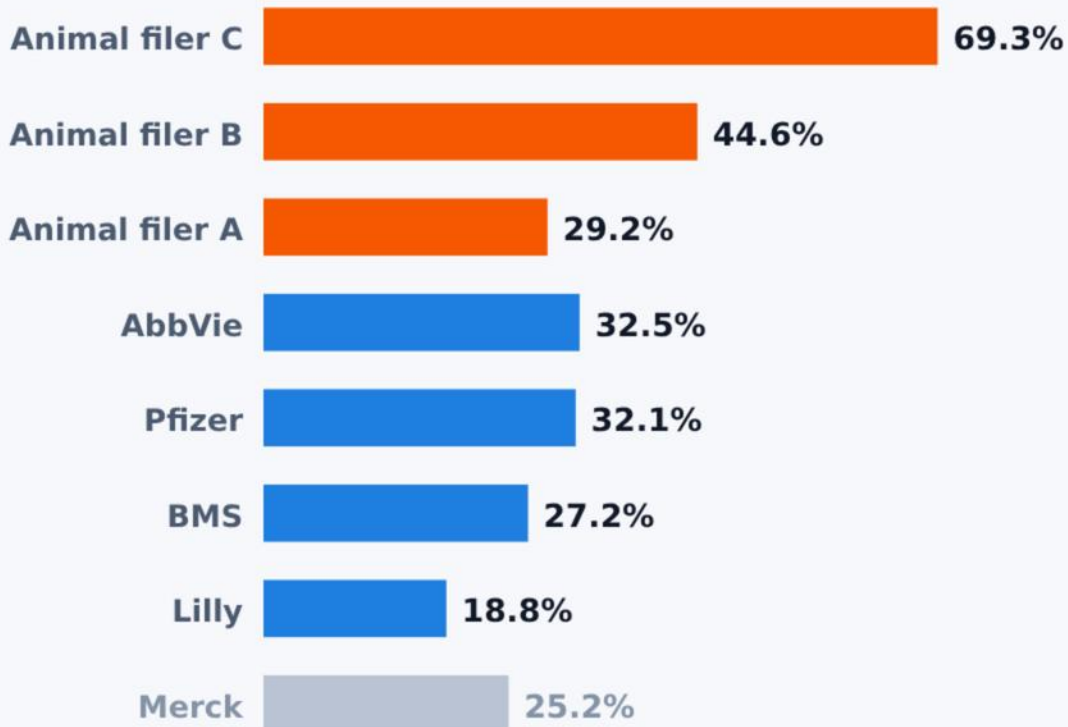
R&D as a share of revenue, FY2023–25 average. Every human filer sits between 17% and 24%. Every animal filer sits **below 8%**.



SEC 10-K FILINGS, FY2023–25 AVG • MERCK'S 34.3% INCLUDES LARGE ACQUIRED-IPR&D CHARGES – ONE REASON IT POOLS WITH NEITHER GROUP

Cost of goods: the ranking inverts.

Cost of goods as a share of revenue. Animal health runs **29–69%**; human pharma 19–33%. **A high COGS share does not mean costs rose — it means price sits close to cost. This is consistent with four decades of veterinary prices tracking inflation.**

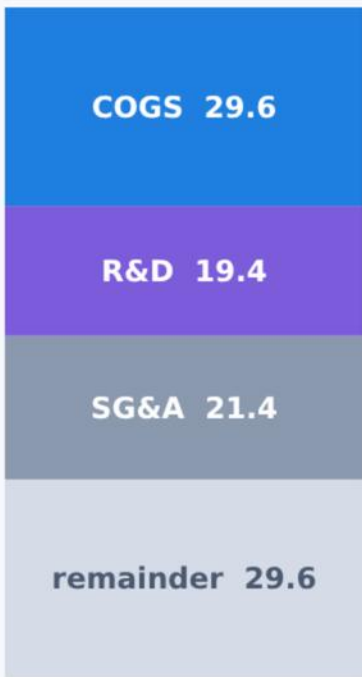


SEC 10-K FILINGS, FY2023–25 AVG • SPREAD INSIDE ANIMAL HEALTH IS REAL: 29% TO 69% ACROSS THE THREE FILERS — PETCARE AND NUTRITION ARE DIFFERENT BUSINESSES

Cost structure per \$100 of revenue.

Group medians, side by side. **Different structures** — and a similar remainder at the end of the line.

HUMAN PHARMA • MEDIAN OF 4



ANIMAL HEALTH • MEDIAN OF 3



MEDIANS: HUMAN N=4, ANIMAL N=3 • "REMAINDER" = 100 - COGS - R&D - SG&A: OPERATING CUSHION BEFORE OTHER LINES, NOT PROFIT

Two cost structures, **two price histories.**

R&D-BUILT P&L • HUMAN

+75%

Real change in the US **producer (factory-gate) price** of human-dominated pharma over the clean 20-year window (2001–2021).

R&D is financed through price — current revenue funds future products.

MANUFACTURING-BUILT P&L • VETERINARY

−4%

Real change in the US **veterinary producer price** on the same window (BLS WPU0634).

Manufacturing follows its input costs — and those inputs became cheaper. Factory-gate, not what clinics or owners pay.

For anyone applying human-pharma benchmarks to animal health: the benchmark does not transfer. Veterinary prices have followed a manufacturing cost base for forty years. Both figures come from public data and both hold after inflation adjustment.

Scope and limitations.

Small samples, stated

Four human filers, three animal. Medians, not means; three fiscal years averaged. This is an anatomy sketch, not a census.

Global, not US-only

10-K figures are worldwide consolidated. Nothing here is a US-market cost share.

Heterogeneity is real

Inside animal health, COGS runs 29% to 69% across the three filers — petcare and nutrition are different businesses sharing a label.

Accounting seams

Filers switched revenue/R&D tags mid-history (unioned, logged); acquired-IPR&D inflates some R&D lines (Merck). No headcount tag exists in XBRL — employee costs stay unmeasured.

Same ingredient market, different businesses.

WHY HUMAN R&D IS HIGH

The pipeline race.

Exclusivity expires after roughly a decade, so the next molecule must always be in trials — and human trials are long, costly and mostly fail. A fifth of revenue goes back in.

WHY ANIMAL COGS IS HIGH

Price near cost, and scale. Prices have tracked inflation for four decades, so cost is a large share of price. Within animal health the share also falls with size — 69% at the smallest filer, 29% at the largest.

THE CONCLUSION

Human pharma sells innovation — priced ahead of cost to fund what's next.
Animal health sells production — priced near cost, earning through volume and efficiency. So: benchmark within the segment.

Every source

Filings · SEC EDGAR XBRL company facts (data.sec.gov), 10-K FY values, FY2023–25: Pfizer, Eli Lilly, AbbVie, Bristol Myers Squibb · Zoetis, Elanco, Phibro · Merck (shown, not pooled). Concept tags per company recorded in the repository.

Price histories · BLS PPIs via FRED — veterinary preparations WPU0634, pharmaceutical preparation industry PCU325412325412, CPI-U deflated, continuous-segment analysis.

Part 1: the input ledger — 44 years of the active ingredient. Every figure in both posts is recomputed from data committed in the repository, and each carries its named source.